

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

Degree, Semester & Branch: B.E., IV Semester, Electrical and Electronics Engineering

Course Code & Title: EE8403 Measurements and Instrumentation

Name of the Faculty member: Ms.S.Sharmila Kumari

Name of the Topic: AC and DC Bridges, PMMC, PMMI, Electrodynamometer, Resistive, Inductive and Capacitive Transducer.

Name of the Innovative Practice: NPTEL Videos, Animation Videos, Solving GATE Questions

Date & Duration: 20.12.18 (50 Minutes), 19.01.19(50 Minutes), 28.01.19(50 Minutes) 14.02.19(50 Minutes), 02.03.19(50 Minutes).

ICT Tool Used: NPTEL Videos, GATE Academy Website

Description:

For better understanding the following NPTEL Videos are shown in class room:

- Functional elements of an instrument (NPTEL- Measurement system characteristics – Dr. S. Mukhopadhyay – IIT Kharagpur -Lecture No.:3)
- Instrument transformer (NPTEL- Potential and Current Transformer –Dr. D. Kastha- IIT Kharagpur - Lecture No.:20)
- D. C. Bridges (NPTEL- Various Resistance Measurement Techniques) –Dr. M. K. Gunashekar- IIS - Lecture No.:29)
- Cathode Ray Tube display (NPTEL- CRT display devices - Dr. Sukhendu Das – IIT. Madras - Lecture No.:02)
- Inductive transducers (NPTEL- Linear Variable Differential Transducers) - Dr. A. Barua – IIT. Kharagpur - Lecture No.:10)

For getting clear idea the following animation videos are shown in class room:

- Analog meters (MC, MI, Electrodynamometer) construction and working principle.
- Types of display devices, working and construction.
- Working of Resistive, Inductive and Capacitive Transducer.

For knowing the importance of the subjects, GATE questions were solved under the following topics:

- AC and DC Bridges, PMMC, PMMI, Electrodynamometer.
- Resistive, Inductive and Capacitive Transducer.

Goals (Learning Outcomes):

- ❖ Acquire knowledge on basic functional elements of instruments and various types of errors present in measurements.
- ❖ Explain the various concepts of electrical and electronics instruments.
- ❖ Compare a suitable measuring instrument used for measuring different electrical quantities.
- ❖ Explain the operating principles of various storage and display devices.
- ❖ Explain the operational features of transducer.

Use of appropriate methods:

Justification for Choosing the Activity for the topic:

➤ **Playing NPTEL Videos and Animation videos in class hour**

After showing NPTEL Videos, the students were able to understand the basics of concepts and as a teacher I will create awareness about NPTEL courses for good learners.

➤ **Solving GATE questions**

After solving GATE questions for particular topics on “AC and DC Bridges, PMMC, PMMI, electrodymanometer, Resistive, Inductive and Capacitive Transducer” the students were able to understand the basics of the concepts and also they know the importance of that particular topics. So they prepare well for examination and I will create awareness about GATE Exam for good learners.

Time Allotted for the Activity: 5 class hours for NPTEL Videos and Animation videos, 10 Minutes for Solving GATE questions

Effective presentation

Details of the Implementation:

After completing the topic, I will project the videos and solving related questions from GATE academy.

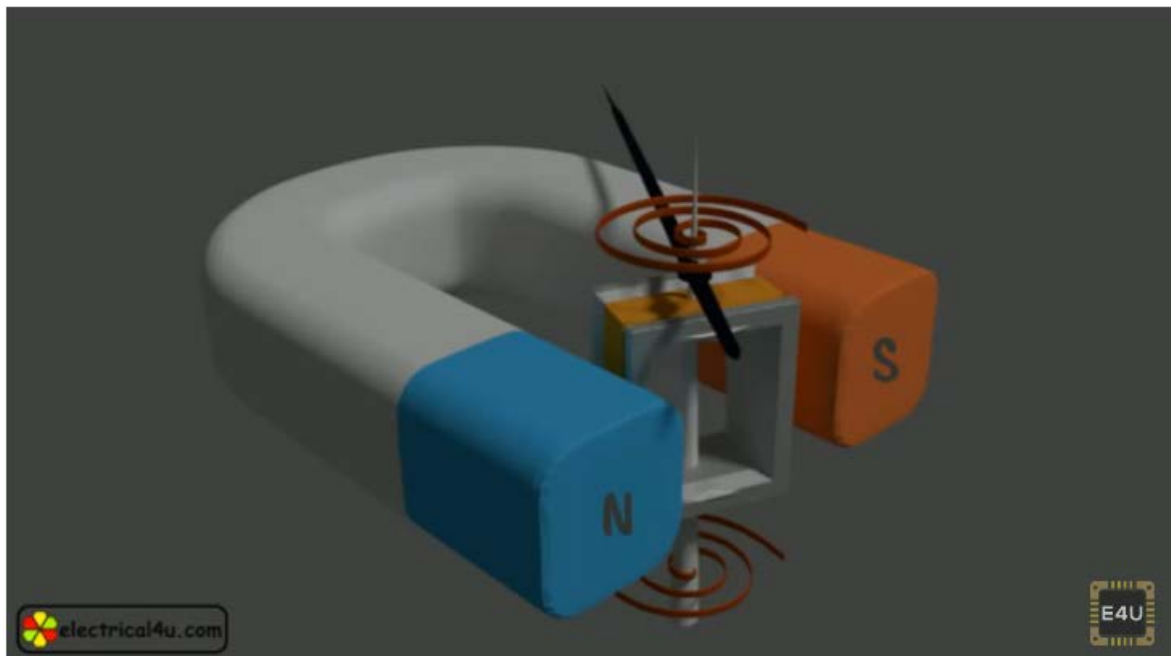
Total Strength is 56; Number of Sessions – Respective Class Hour (50 Minutes)

Photographer: Snapshot from NPTEL and Animation Videos

Reporter: Myself

Session detail

- ✓ I explained the students about the respected unit for 15 minutes.
- ✓ Then, I will project the videos from respective websites for 15 minutes
- ✓ Then, I will solve the GATE questions in whiteboard and asked the question in class hours.
- ✓ Finally, I summarized the points told by the students.



Snapshot form Animation Videos

- Slow learners were not able to understand some standard formulas and topics during GATE questions discussion hours.
- I need 5 more hours in lecture plan for conducting this activity.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
1	3	3	1	-	1	-	-	-	-	1	-	1
2	3	2	2	-	1	-	-	-	-	1	-	1
3	2	2	3	-	1	-	-	-	-	1	-	1
4	3	2	3	-	1	-	-	-	-	1	-	1
5	2	1	2	-	1	-	-	-	-	1	-	1

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. H.S. Kalsi, 'Electronic Instrumentation', McGraw Hill, III Edition 2010.
3. Functional elements of an instrument (NPTEL- Measurement system characteristics – Dr. S. Mukhopadhyay – IIT Kharagpur -Lecture No.:3)
4. Instrument transformer (NPTEL- Potential and Current Transformer –Dr. D. Kastha- IIT Kharagpur - Lecture No.:20)
5. D. C. Bridges (NPTEL- Various Resistance Measurement Techniques) –Dr. M. K. Gunashekar- IIS - Lecture No.:29)
6. Cathode Ray Tube display (NPTEL- CRT display devices - Dr. Sukhendu Das – IIT. Madras - Lecture No.:02)
7. Inductive transducers (NPTEL- Linear Variable Differential Transducers) - Dr. A. Barua – IIT. Kharagpur - Lecture No.:10) <http://nptel.iitm.ac.in>.